

Week 3

3.3 | Modelling, Analysis and Recommendation



⌚ 30 minutes

The modelling stage of the Team Data Science Process (TDSP) aims to achieve the following goals:

- **Determine Optimal Data Features:** Identify the most relevant data features for the machine-learning model.
- **Create an Informative Model:** Develop a machine-learning model that accurately predicts the target variable.
- **Prepare for Production:** Ensure that the machine-learning model is suitable for deployment in a production environment.

The key tasks in this stage include feature engineering, model training, model evaluation, interpreting the model, and assessing fairness. Feature engineering involves creating data features from raw data to facilitate model training. Model training includes building and evaluating models to find the best solution. Model evaluation focuses on assessing the model's performance and making decisions about its suitability for production. Interpreting the model involves understanding the model's behaviour and individual predictions, while assessing fairness aims to evaluate the fairness of the model's predictions.

The modelling stage is a critical phase in the TDSP lifecycle, where data scientists work on developing and refining machine-learning models to address specific business questions and requirements.

The challenge

You will apply logistics regression to build predictive model for the dataset explored in the Challenge 3.1 and Challenge 3.2. You will need to evaluate the performance of the predictive model using a test set (we will have more in-depth

discussion of evaluation metrics in the upcoming sections). You will apply this to the fictional company ConnectTel.

The output



A simple machine learning model for churn prediction and completion of the 'Modelling' and 'Metrics' and 'Analyses and Recommendations' sections in the 'ConnectTel Project Documentation Template'.

The task

Emily is very happy with your work so far and think that your insights align well with her understanding so far. She thinks that you are ready for the next challenge: modelling. In this task, you'll use the Logistics regression to complete the 'Modelling' sections of the ConnectTel template.

Project conversations

Review the following conversations to gain a deeper understanding of the role of modelling in data science.

Quote text

You (Project Manager): Hey Emily, can you explain what modelling in data science, in a simple way?

Emily (Data Scientist): Sure! Think of it like baking a cake. First, we gather our ingredients (data), making sure they're fresh and high-quality. Then, we mix them together in the right proportions—that's our algorithm. Just like in baking, the temperature and time matter; we adjust these (our model parameters) until the cake (our model) comes out just right!

You: So, it's a bit of art and science?

Emily: Exactly! We need the right ingredients (data), the right recipe (algorithm), and a bit of intuition and experience to make it perfect. Like a master baker, a data scientist knows how to blend these elements to create something delicious, or in our case, a model that really works!

You: And how do we know if our cake is a hit?

Emily: We taste it! Or in data science, we test our model with new data to see how well it performs. If it's not quite right, we might tweak the recipe or bake it a little longer. It's all about experimenting and fine-tuning until we get that perfect cake—I mean, model!



Instructions

Review the task materials, then work through the following steps to create the output you will need for the 'Model Evaluation Metrics' and 'Analyses and Recommendations' sections of Assessment 1. Select the arrow to expand the instructions:

▼ Building ML model in Google Colab with AI Agent

Emily suggested that you try Logistics Regression, a classic algorithm but usually used to a create baseline for predictive modelling. Explore logistics regression, training and validation in the supporting resources to complete the 'Model Selection Rationale' and 'Model Training and Validation' section in the ConnectTel template.

In the following task instructions, to simplify the pipeline, the same dataset will be used for training and validation.

Emily suggested the follow steps to build the Logistics regression model with **Google Colab with AI Agent**:

1. Continue working with the notebook in Google Colab created in [3.2 | Data Acquisition and Understanding](#).
2. Design a prompt for this task, e.g. "Help me develop a logistics regression model for churn prediction where <variable_name> is the target variable. Randomly split the dataset: 60% is used for training and 40% is used for testing,. Evaluate the performance of the model using accuracy".
3. Review the plan proposed by the AI agent - edit the plan to incorporate your insights.
4. Optional: Develop another model based on decision trees and compare the performance of the two models.
5. Use the results and analyses here to complete the 'Model Evaluation Metrics' and 'Analyses and Recommendations' sections in the ConnectTel Project Documentation Template.



Come prepared to discuss your insights and your recommended performance model in the live class. Start planning for the modelling and analysis sessions for the Modern Furniture Co. case study.

Supporting resources

- The following resource explains logistics regression: [Understanding Logistic Regression for Machine Learning Classification \(Part 14 of 17\) | Machine Learning for Beginners](#) ➞ (Microsoft Learn, 2023)
- The following resource explains training sets, validation sets, and test sets: [Why do we split data into train test and validation sets?](#) ➞ (Turp, 2023)



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